

RESEARCH ARTICLE

Hydroxyalkylation of Benzylic C—H Bonds via Heterogeneous Photocatalysis

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ABSTRACT

Hydroxyalkylation of benzylic C—H bonds with alcohol can not only construct C(sp³)—C(sp³) bonds but also introduce hydroxyl groups into the organic scaffold, which offers an efficient and straightforward approach to access versatile benzylic moieties. Here we disclose a protocol of hydroxyalkylation of benzylic C—H bonds enabled by the hydrogen bonding-assisted cage effect (HBACE). A broad scope of benzylic C—H substrates can be converted into desired products with excellent selectivity. The transformation follows a radical cross-coupling mechanism which may be dominated by the persistent radical effect.

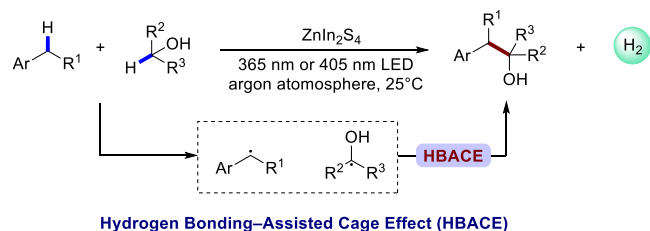
Benzylic moieties are significant and ubiquitous fragments in fine chemicals and pharmaceuticals, developing strategies for access to versatile benzylic moieties in a step and atom economic manner is highly desirable. Direct functionalization of benzylic C—H bonds has been considered as an efficient and straightforward strategy for access to benzylic moieties, and it has attracted increasing attention in the past decades [1–9].

Functionalization of benzylic C—H bonds using readily available alcohols is very promising and synthetically useful. One example is alkoxylation of benzylic C—H bonds with alcohol, which has been well established [4–5, 8]. Another extremely attracting transformation is hydroxyalkylation of benzylic C—H bonds with alcohol via a cross-coupling dehydrogenation reaction. This transformation can not only construct C(sp³)—C(sp³) bonds but also introduce hydroxyl groups into the organic scaffold, which is of great significance for synthetic chemistry. However, such a transformation is very challenging, it requires simultaneous cleavage of both the benzylic C(sp³)—H and the C(sp³)—H of alcohols in one catalytic system, moreover, a subsequent cross-coupling reaction along with the evolution of hydrogen gas must be realized.

Herein, hydroxyalkylation of benzylic C—H bonds with alcohol via a radical cross-coupling reaction by heterogeneous photocatalysis is disclosed (Scheme 1). In this strategy, the hydrogen bonding-assisted cage effect (HBACE) plays a key role in achieving high cross-coupling selectivity with respect to benzylic substrates. To the best of our knowledge, this is the first example which realizes hydroxyalkylation of benzylic C—H bonds with alcohol.

Coupling of toluene and propan-2-ol was selected as a model reaction, and ZnIn₂S₄ (i.e. ZIS-A and ZIS-B) [10] with a hexagonal phase (Figure S1) was employed as a photocatalyst. The reaction conditions were optimized via a series of control experiments (Table 1). A good yield of 52% for the desired product with excellent selectivity was obtained (Table 1, Entry 1) under standard reaction conditions, and no homo-coupling product of toluene was detected.

It was found that solvents play a key role in current transformation and that *t*-butanol provides the best yield (Table S1). We then examined the influence of the equivalent of propan-2-ol on the reaction using *t*-butanol as a solvent (Table S2). The yield of **1** increased when the equivalent of propan-2-ol was



SCHEME 1 | Hydroxyalkylation of benzylic C–H bonds with alcohol enabled by the HBACE.

increased (Table S2, Entries 1–6), and propan-2-ol/*t*-butanol (3:1 v/v) provides similar yield as the neat condition. The reaction performed in the absence of acetic acid furnishes a slightly lower yield (Table 1, Entry 2), and optimization of the equivalent of acetic acid was also conducted in which 2 equivalents of acetic acid was chosen as the standard reaction condition (Table S3). When trifluoroacetic acid was used instead of acetic acid, a lower yield was found (Table 1, Entry 3). When the reaction was performed at 60°C, a minimal change of yield was found whereas the reaction rate was increased (Table 1, Entry 4). ZIS-B as a photocatalyst was also examined, however, a much lower yield (20%) (Table 1, Entry 5) was obtained. The difference in photocatalytic activity for these two photocatalysts may be attributed to that the ZIS-A has better separation and transfer of photogenerated carriers [10]. Interestingly, when 365 nm LED was employed, the reaction rate was increased and a similar yield (55%) was obtained (Table 1, Entry 6). The higher reaction rate could be attributed to a larger absorbance at 365 nm compared with that at 405 nm for the ZIS-A (see Figure S2). The reaction conducted without degassing provides a lower yield (41%, Table 1, Entry 7). When 10 equiv. of water was added, a slightly lower yield (48%) was obtained (Table 1, Entry 8), indicating that the current protocol is not sensitive to the presence of water. No target

product was detected when the reaction was conducted in dark or in the absence of the photocatalyst (Table 1, Entries 9 and 10).

With the optimized reaction conditions in hand, we then investigated the scope of this protocol in a flow photoreactor (Figure 1). With respect to benzylic partners, the protocol functions well with toluene derivatives (1–17) with moderate to good yields of the desired products. It is worth noting that the cross-coupling selectivity with regard to benzylic substrates is good to excellent (67–92%). In the presence of more than one benzylic C–H site within a substrate, monofunctionalized products were obtained (2–5). A variety functional groups including acetal (7), acetoxy (8 and 17), carboxyl (9), halogen (10–13), trifluoromethyl (14), trifluoromethoxy groups (15) are tolerated. Compared with a substituent at *para*-position, the same substituent at *meta*-position provides higher yield (3 and 4; 10 and 11). To our delight, pinacol phenylboronic ester, which is a coupling partner in the Suzuki–Miyaura coupling reaction [11], is compatible with the present protocol (16), and this methodology offers an attractive alternative for access to hydroxy-containing phenylboronic esters.

Substrates with secondary benzylic C–H sites are also amenable to the transformation (18–24). Interestingly, in the presence both primary and secondary benzylic C–H sites, a regioselectivity of 3:1 for the secondary position was observed (21). The cleavage of the secondary benzylic C–H could be faster than that of the primary one, which may dominate the regioselectivity. Next, we examined the scope of this protocol with regard to alcohol. When diphenylmethane was employed as the benzylic C–H substrate, methanol, ethanol and propan-1-ol furnish the desired products with moderate yields (25–27). Further optimization of these yields via precisely controlling formation rates of two radicals is possible (see discussion below). An application of current methodology in the late-stage functionalization was also

TABLE 1 | Optimization studies for hydroxyalkylation of benzylic C–H bonds with propan-2-ol^a.

Entry	Deviation from standard conditions	Conversion ^b %	Yield ^b %	Selectivity ^c %
1	None	59	52	89
2	Without acetic acid	51	45	87
3	CF ₃ COOH (2 equiv.)	54	44	82
4	60°C, 12 h	60	50	85
5	ZIS-B instead of ZIS-A	37	20	53
6	365 nm LED (5 W), 12 h	63	55	87
7	Without degassing	69	41	59
8	Add 10 equiv. H ₂ O	56	48	86
9	Without light	8	ND	NA
10	Without ZIS-A	3	ND	NA

Abbreviations: ND: not detected. NA: not applicable.

^aZIS-A (10 mg), toluene (0.2 mmol, 50 mM), acetic acid (0.4 mmol, 2 equiv.), propan-2-ol (4 mL), Ar, 25°C, 405 nm LED (5 W), 24 h.

^bConversion of toluene and yield of the product was determined by HPLC.

^cSelectivity: yield of the product divided by conversion of toluene.

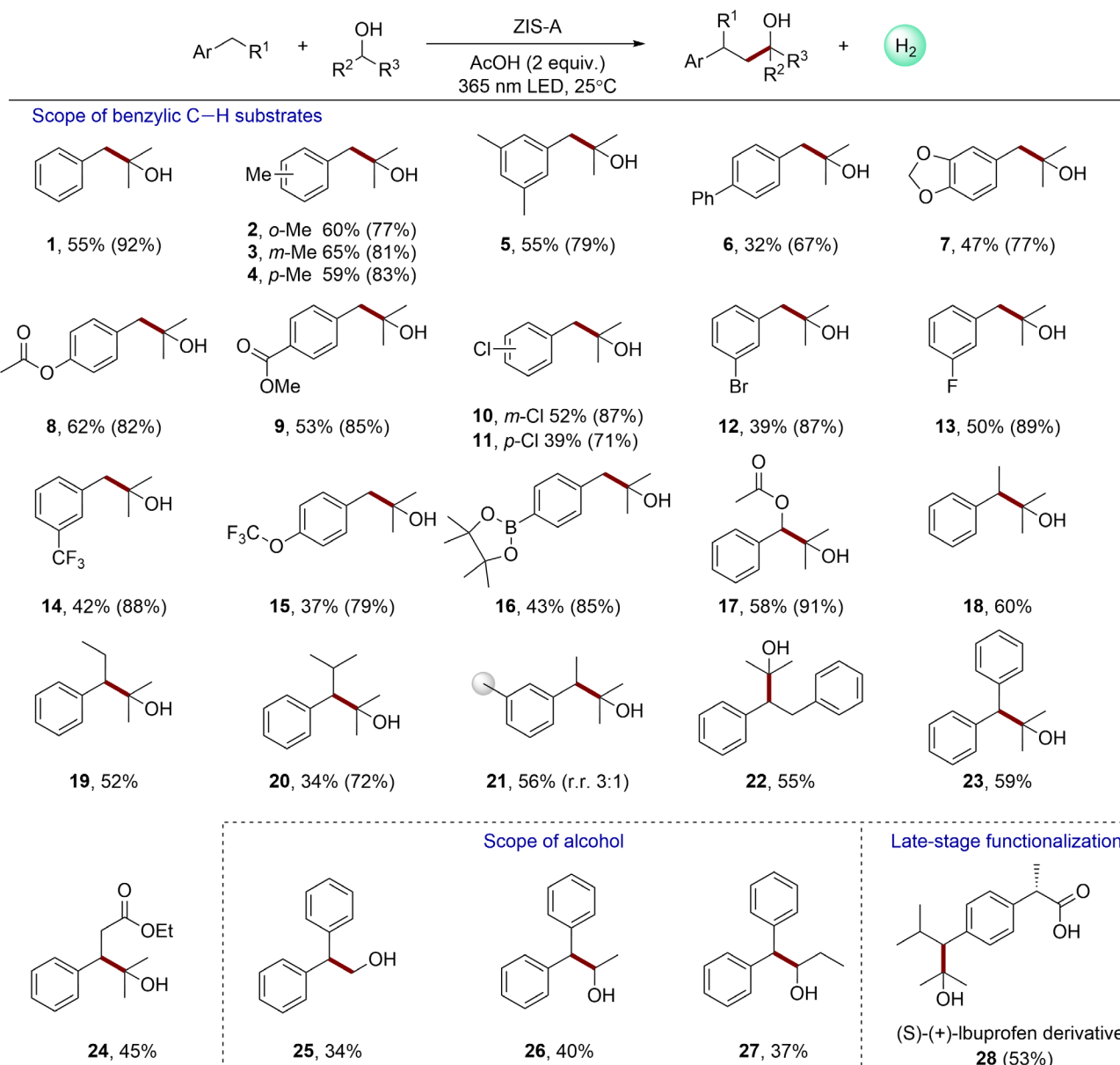


FIGURE 1 | Scope of the protocol. Reaction conditions: typically, ZIS-A (35 mg), benzylic C–H substrates (1.75 mmol), alcohol (35 mL), Ar, 25°C, 365 nm LED (40 W), see the Supporting Information for experimental details. Isolated yields (yields shown in parentheses were based on the consumed benzylic C–H substrates).

investigated, a derivative of ibuprofen (**28**) could be obtained with good yield, demonstrating a potential application of current protocol in the drug diversification.

The reaction mechanism was investigated by a series of experiments.

First, we performed the wavelength-dependent experiments in which photon flux was kept the same for different wavelength (see Table S4). As shown in Figure 2A, yields of **1** under irradiation with different wavelength corresponds well with the absorbance of the ZIS-A, which reveals that the reaction was driven by the photocatalyst. An intermittent irradiation experiment (Figure 2B) indicates that a continuous irradiation was essential for the formation of **1**, excluding the possibility of a radical chain mechanism.

We then investigated the mechanism for activation of the two types of C(sp³)–H bonds by the ZIS-A. The activation of α-C(sp³)–H of alcohols by thiyl radicals on the surface of the ZIS-A has been reported [12]. Given a smaller bond dissociation energy (BDE) of benzylic C(sp³)–H (89 kcal/mol for toluene) compared with that of α-C(sp³)–H of propan-2-ol (95 kcal/mol) [13], it is reasonable to speculate that the ZIS-A is capable of activating the benzylic C(sp³)–H. The existence of the radicals during the reaction was investigated by in situ electron paramagnetic resonance (EPR) measurements in which 5,5-dimethyl-1-pyrroline-*N*-oxide (DMPO) was used as a spin-trapping agent. As shown in Figure 2C, a six-line EPR signal ($a_N = 14.8$ G, $a_H = 23.8$ G) attributed to a signal of an adduct formed by the 2-hydroxypropyl radical and the DMPO was detected after irradiation [14]; when the reaction was conducted in the absence of toluene, a similar EPR signal was observed (see Figure S3),

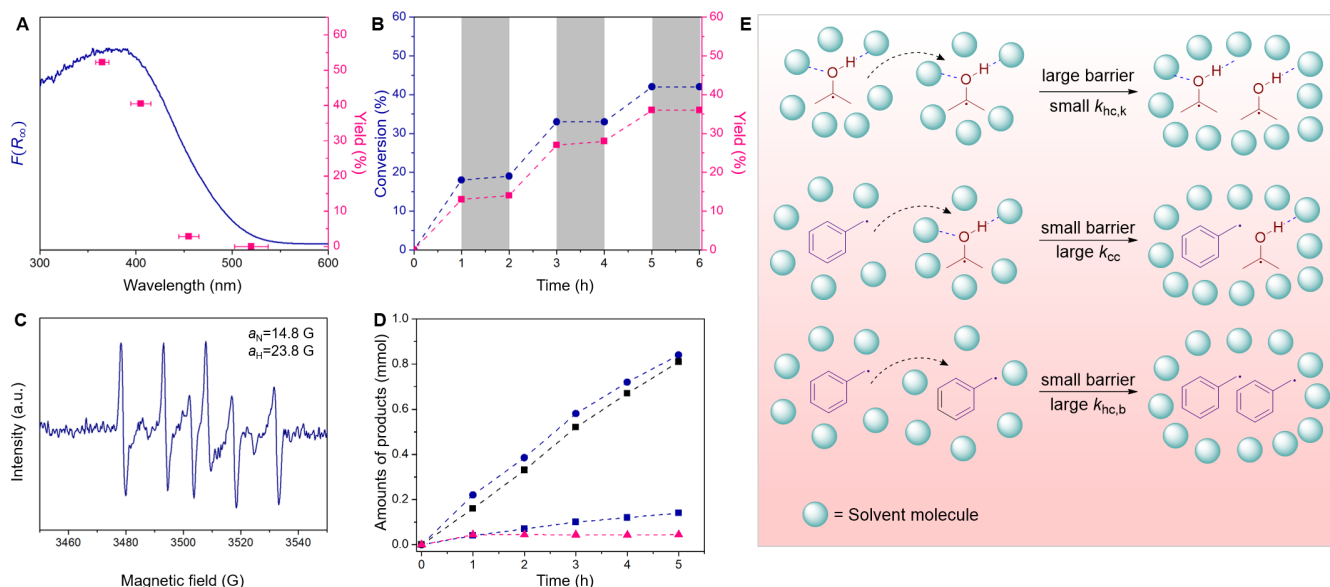


FIGURE 2 | Mechanistic studies. (A) Wavelength-dependent experiments. (B) Intermittent irradiation experiment. Gray area represents period during which light is off. (C) An in situ EPR spectrum using DMPO as a spin-trap agent. (D) Amounts of **1** (circle), pinacol (square) and 1,2-diphenylethane (triangle) during reactions conducted in the absence of toluene (black trace), in the absence of propan-2-ol (red trace) and under the standard conditions (blue trace), respectively. (E) A schematic illustration of the hydrogen bonding-assisted cage effect which dominates the cross-coupling selectivity.

indicating the existence of 2-hydroxypropyl radicals during the reaction. Notably, signals of benzyl radicals trapped by the DMPO were not observed in Figure 2C, possibly owing to the lower concentration of the benzyl radicals compared with that of the 2-hydroxypropyl radicals during the reaction (see the discussion of the kinetics of radicals below). When S species on the surface of the ZIS-A was benzylated, the photocatalyst was almost inactive for the reaction (see Table S5), signifying that thiyl radicals on the surface of the ZIS-A are active sites for activation of the two types of $C(sp^3)-H$ bonds.

To elucidate kinetics regarding the formation and the transformation of the radicals during the reaction, control experiments were conducted in the absence of toluene, in the absence of propan-2-ol and under the standard conditions, respectively (Figure 2D). Amounts of the products as a function of the irradiation time were measured, when the reaction was performed in the absence of toluene, pinacol which is a homocoupling product of propan-2-ol was found and isolated (see the Supporting Information). When the reaction was conducted in the absence of propan-2-ol (*t*-butanol as solvent), a homocoupling product of toluene was detected whereas the formation of the homocoupling product ceased after 1 h. This could be ascribed to that thiyl radicals which act as active sites for the $C-H$ activation were consumed by the benzyl radicals via $C-S$ bond formation. Fourier transform infrared (FTIR) spectrum of the recycled ZIS-A (Figure S4) demonstrates a characteristic band at 697 cm^{-1} which can be attributed to out-of-plane $C-H$ bending of benzene ring or $C-S$ stretch [12]. The detection of the corresponding homocoupling products further confirms the existence of two types of radicals during the photocatalytic reaction. When the reaction was performed under the standard reaction conditions, the cross-coupling product was obtained with high selectivity and no homocoupling product of

toluene was detected (yield lower than 2% based on limit of detection).

The high cross-coupling selectivity with respect to benzylic substrates was then investigated. According to the persistent radical effects (PRE) [15, 16], high selectivity could be realized when the two types of radicals have equal formation rates and have different rate constants for their homocoupling reactions. These two criteria could be satisfied in current catalytic system. For the hydroxyalkylation of benzylic $C-H$ bonds of toluene with propan-2-ol, given a smaller BDE of benzylic $C(sp^3)-H$ (89 kcal/mol) compared with that of $\alpha-C(sp^3)-H$ of propan-2-ol (95 kcal/mol) [13], The rate constant for cleavage of benzylic $C(sp^3)-H$ could be larger than that for cleavage of $\alpha-C(sp^3)-H$ of propan-2-ol. On the other hand, the concentration of propan-2-ol (solvent-quantity) is much higher than that of benzylic substrate, which compensates the smaller rate constant. Therefore, formation rates of the two types of radicals could be equal.

Benzylic and ketyl radicals are usually regarded as transient radicals, in current system, solvation plays a key role in determining rates of radical coupling (Figure 2E). Ketyl radicals can participate formation of hydrogen bond with the solvent (e.g. propan-2-ol or *t*-butanol) whereas benzyl radicals could not [17]. Therefore, the diffusion barrier of ketyl radicals could be higher than that of benzyl radicals, which results in that ketyl radicals are less transient than benzyl radicals ($k_{hc,k} \ll k_{hc,b}$). It should be noted that the barrier for the diffusion of the benzyl radical into the cage surrounding a ketyl radical or the cage surrounding a benzyl radical could be small, and the rate constant of the cross-coupling reaction may have same magnitude as that of homocoupling of benzyl radicals ($k_{cc} \approx k_{hc,b}$). Therefore, the ratio of two rate constants of homocoupling reactions ($k_{hc,b}/k_{hc,k}$) for the two radicals could be large enough to satisfy the second requirement.

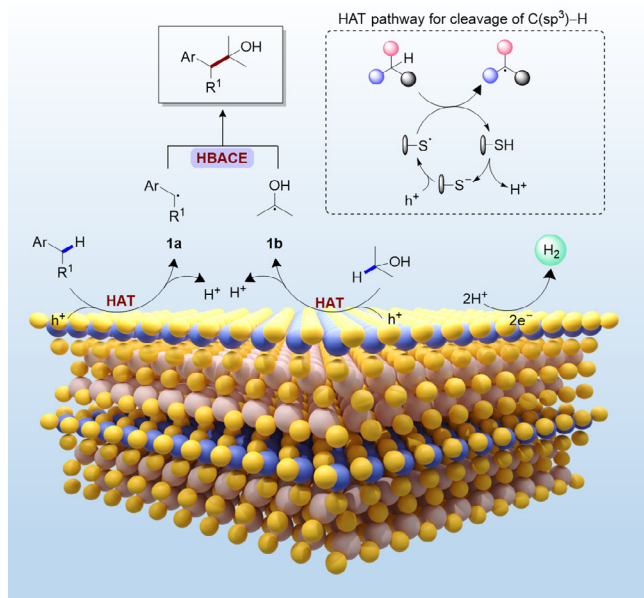


FIGURE 3 | A plausible mechanism for hydroxyalkylation of benzylic C–H bonds with propan-2-ol over the ZIS-A.

For a substrate with secondary benzylic $C(sp^3)$ –H such as diphenylmethane, the formation rate of benzyl radicals could be higher than that for the primary benzylic substrate if the two substrates have same concentration, and employing the same reaction conditions for the primary benzylic $C(sp^3)$ –H only furnishes 29% yield of the target product **23**, further optimizing formation of the benzyl radicals such as decreasing the concentration of the benzylic substrate resulted in a significantly improved yield (58%, see Tables S6 and S7).

A plausible mechanism for the hydroxyalkylation of benzylic C–H bonds with propan-2-ol over the ZIS-A was proposed according to above results. As shown in Figure 3, photogenerated holes could be trapped by S^- (or SH) sites, affording $S^\bullet$ species which enable cleavage of both the benzylic $C(sp^3)$ –H and the α - $C(sp^3)$ –H of alcohol via hydrogen atom transfer (HAT). Subsequently, the benzyl radicals and the ketyl radicals undergo a cross-coupling reaction enabled by the HBACE, affording the target product in high selectivity. Finally, photogenerated electrons reduce protons to hydrogen.

In summary, we have developed a protocol of hydroxyalkylation of benzylic C–H bonds with alcohol for $C(sp^3)$ – $C(sp^3)$ bond formation under extremely mild reaction conditions. A variety of benzylic C–H substrates are well tolerated, and target products were obtained with good to excellent selectivity. Given that the hydroxyl group has various transformations, the products can be regarded as a platform for versatile functionality. We believe that the hydrogen bonding-assisted cage effect (HBACE) could be applied in other radical cross-coupling reactions, offering possibility to discover fantastic transformations.

Acknowledgments

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Conflicts of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: cctc70545-sup-0001-SuppMat.pdf.